EST. 2025

SECURE DEV LABS

SOFTWARE AGENCY

Crafting Secure Digital Experiences

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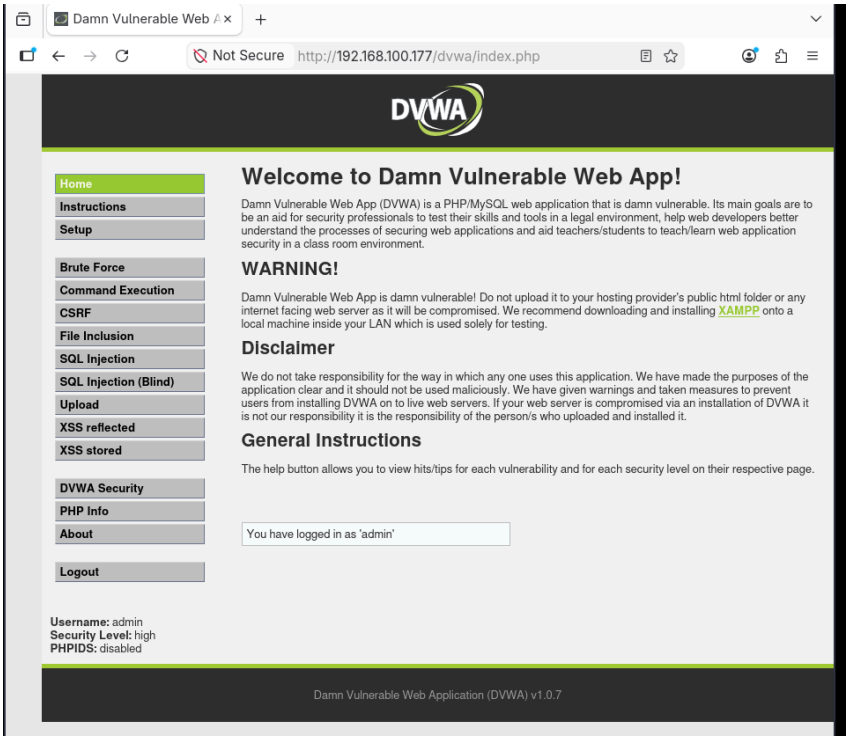
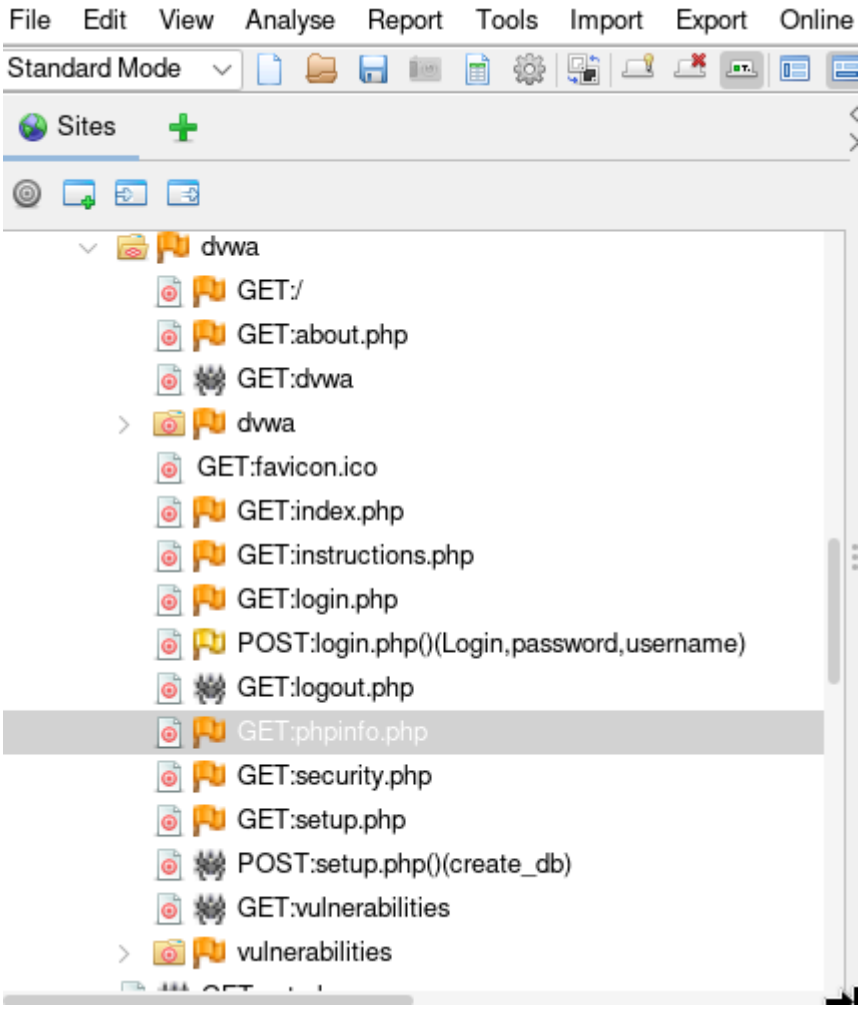
Track	Ethical Hacking	Instructor	Muhammad Saad

Task 1:

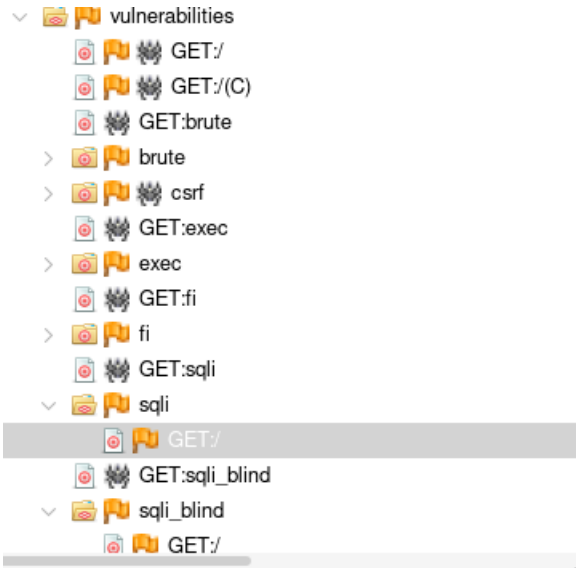
Installation Proof:



For all the endpoints



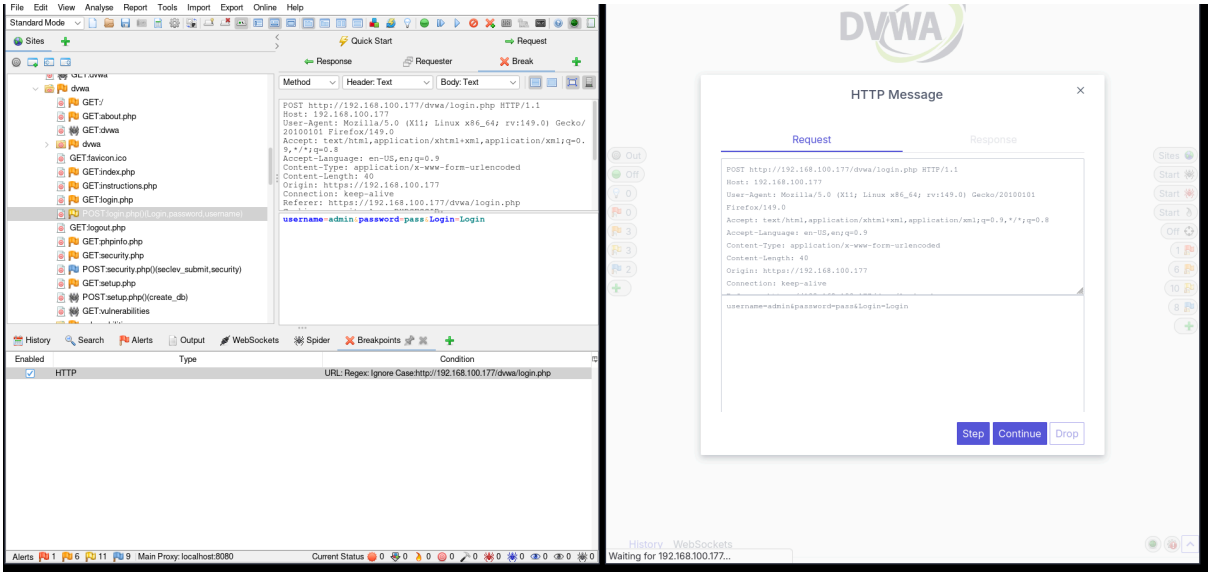
Now the enndpoints which have input fields are all the ones with post
le in login(password,username)
In setup (create_db)



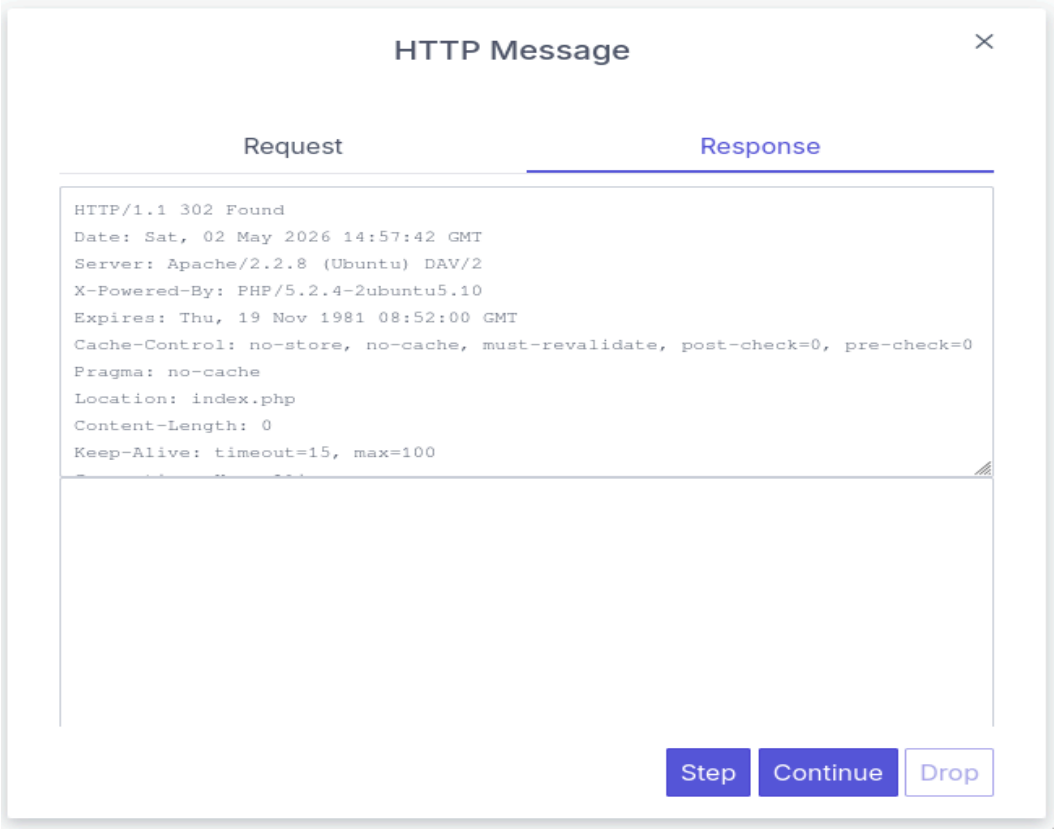
These are all the endpoints in DVWA

Task 2:

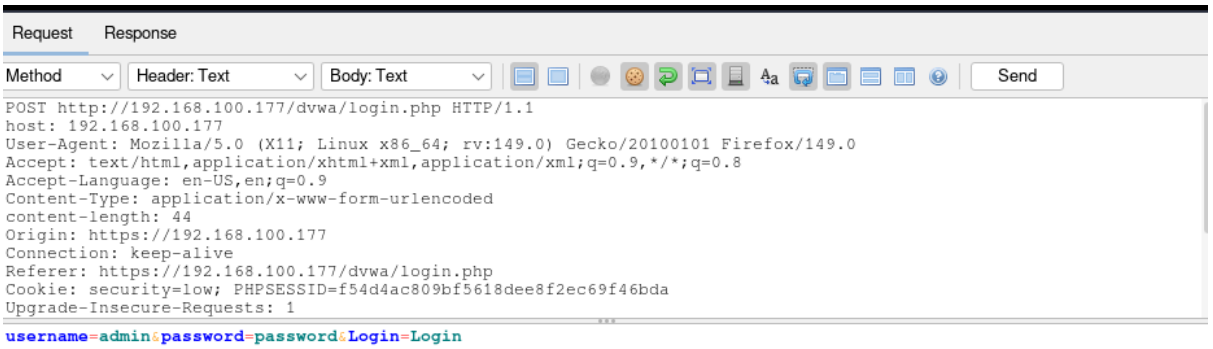
We first set ZAP/Burpsuit to intercept packets and then pressed the login button s that it can capture the request



Then pressed step so that it can be sent forward and we can get a response



And then we accessed the end point through the request editor



Theory:

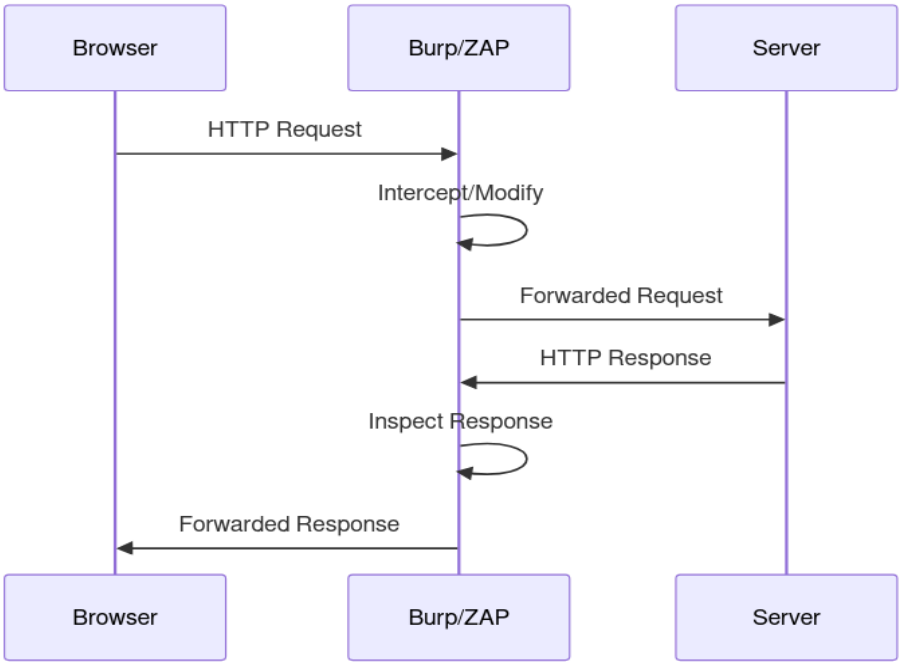
What is Burp Suite and how is it different from Nmap:

Burp Suite is a web application proxy, it sits between your browser and the server, intercepting and allowing modification of HTTP/S traffic. It operates at the **application layer** (Layer 7).

Nmap is a network scanner, it probes hosts for open ports and running services. It operates at the **network/transport layer** (Layers 3-4).

Burp asks: *what is this web app doing?* Nmap asks: *what ports/services are running on this machine?*

Burp/ZAP as an interceptor



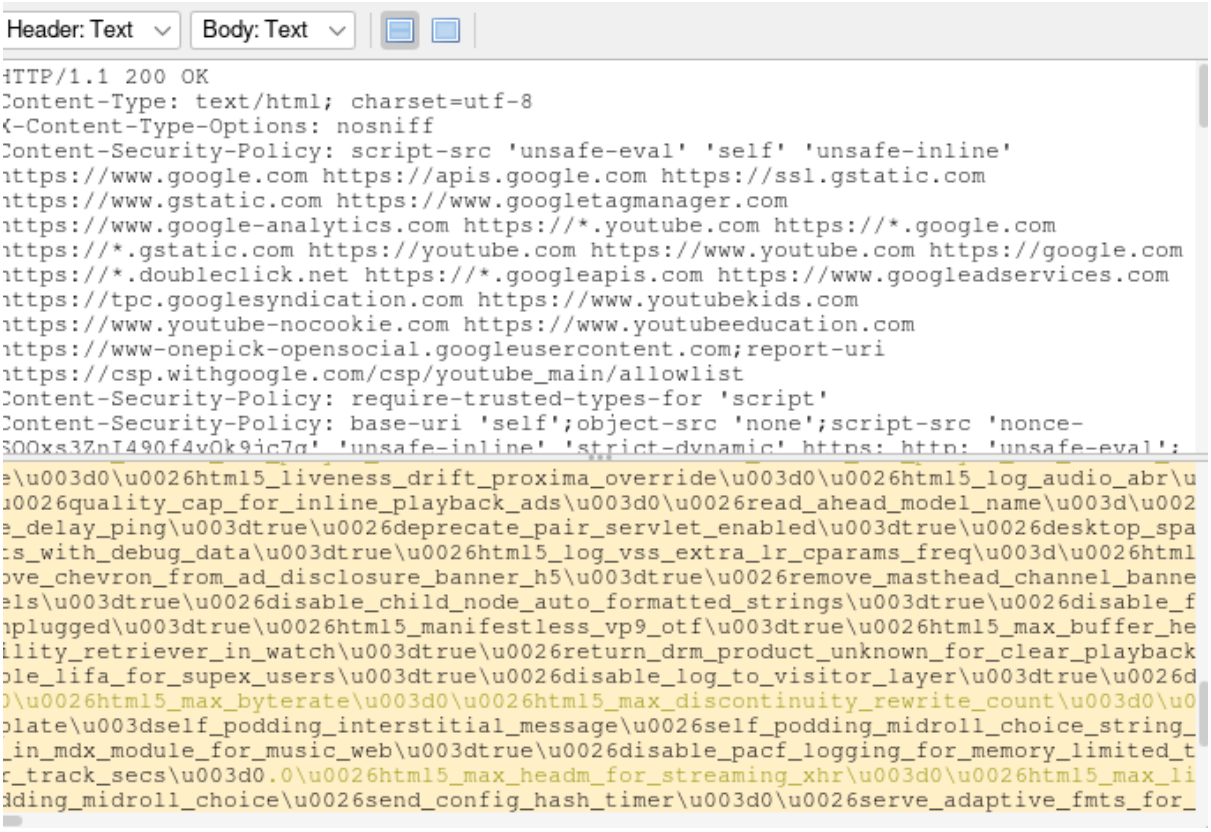
Task 3

Requested Youtube

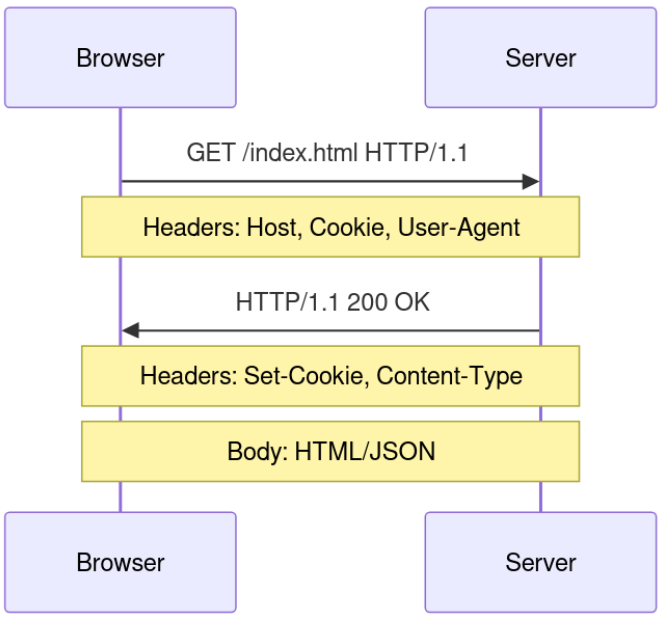
Header: Text | Body: Text

```
GET https://www.youtube.com/ HTTP/1.1
host: www.youtube.com
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:149.0) Gecko/
20100101 Firefox/149.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.
9,*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Upgrade-Insecure-Requests: 1
Service-Worker-Navigation-Preload: true
Connection: keep-alive
Sec-Fetch-Dest: document
Sec-Fetch-Mode: navigate
Sec-Fetch-Site: same-origin
Priority: u=2
```

Response:



Theory



What is Request & Response in web context:

A request is a structured message sent by the browser to a server asking for a resource or action. A response is the server's structured reply containing a status, metadata, and optionally a body.

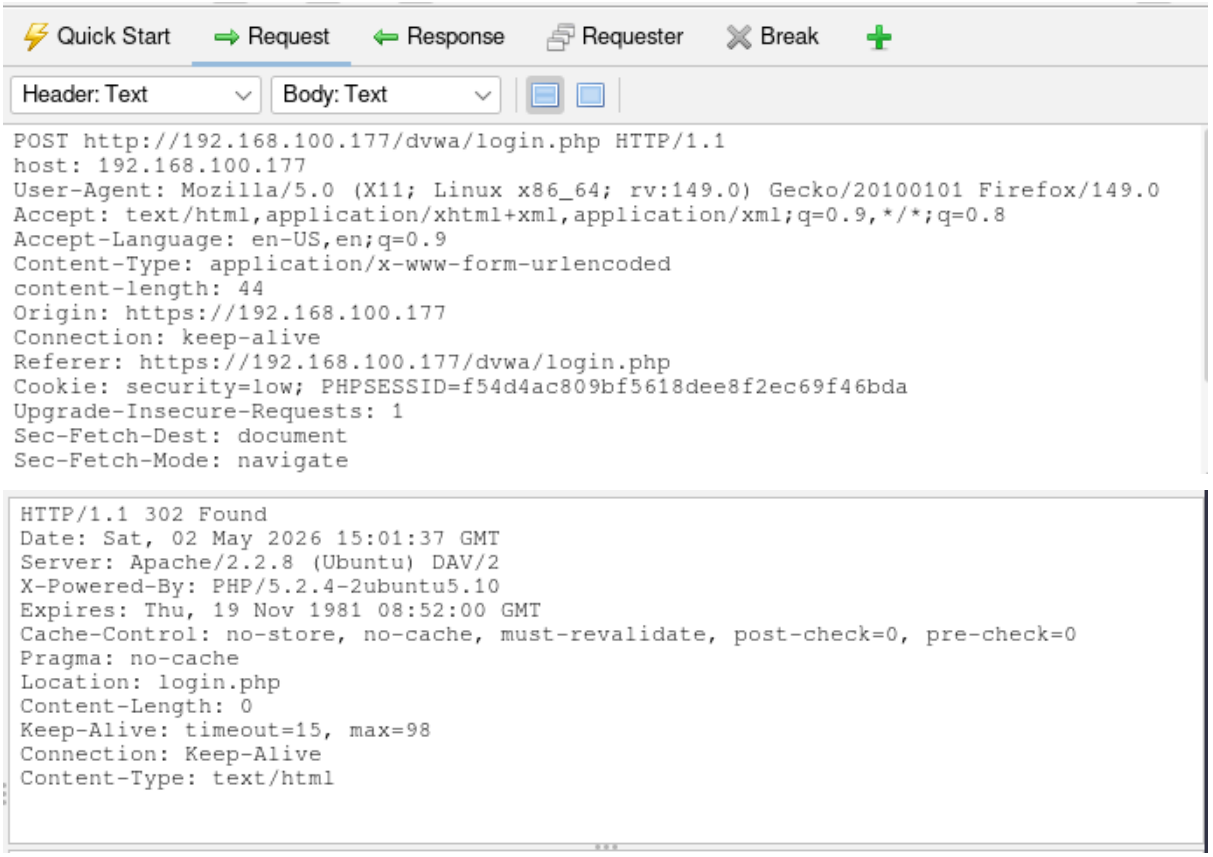
Sender/Receiver analogy:

The browser (sender) writes a letter specifying what it wants (verb, path, headers, body) and mails it. The server (receiver) reads it, processes it, and sends feedback (status code, headers, body) back.

Common HTTP response codes:

- `200` OK, success
- `301/302` Redirect
- `400` Bad request (client error)
- `401` Unauthorized (not authenticated)
- `403` Forbidden (not authorized)
- `404` Not found
- `500` Internal server error

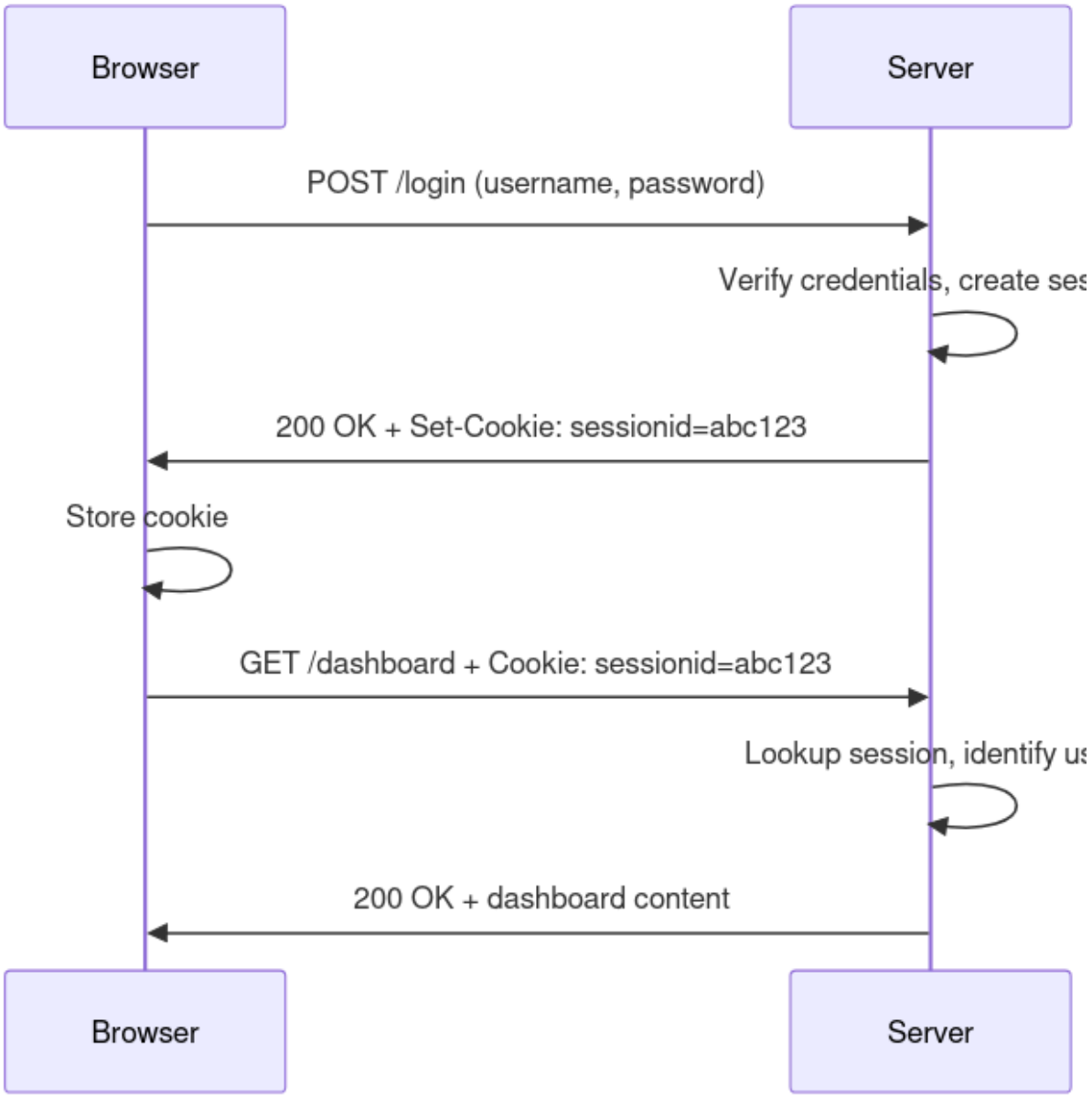
Task 4



The credesentials werenot echoed back by the server

Now for a different directory

Theory:



Are cookies and sessions important for modern web:

Yes, HTTP is stateless by design. Without cookies/sessions every request would be anonymous. They are the only mechanism that gives web apps memory across requests. Every modern auth system depends on them.

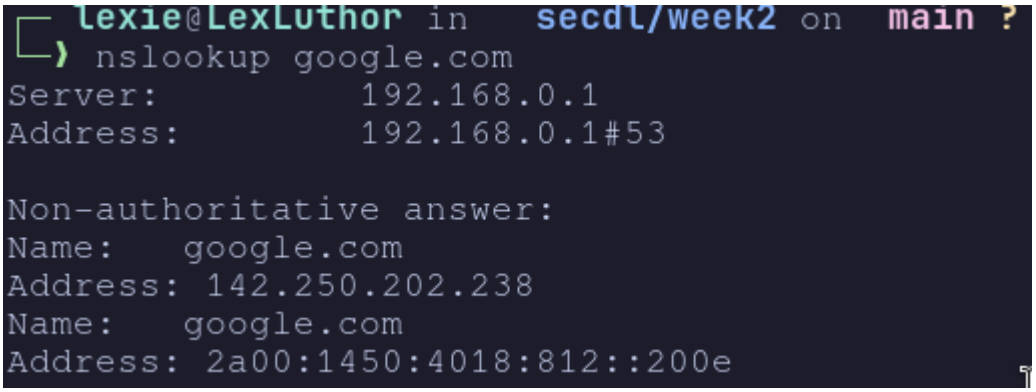
AuthN (Authentication): The process of verifying identity proving you are who you claim to be. Example: submitting username and password.

AuthZ (Authorization): The process of verifying permissions determining what an authenticated identity is allowed to do. Example: checking if you can access `/admin`.

One difference: AuthN happens once at login. AuthZ happens on every single request.

Why cookies/sessions are used: To simulate statefulness on top of a stateless protocol. The server issues a session ID after AuthN, stored in a cookie, which the browser sends automatically on every subsequent request so the server can identify you.

Task 5:



What is DNS: A distributed system that translates human-readable domain names into machine-readable IP addresses. It is the internet's phonebook.

What is a DNS request: A query sent by your machine asking "what IP address does this domain name map to?"

What is a DNS response: The answer returned by a nameserver containing the IP address (or an error if not found).

Domain Vs IP

Domain name	IP address

Domain Vs IP

Human-readable	Machine-readable
`google.com`	`142.250.180.46`
Easy to remember	Required for routing
Can change mapping	Actual network address

Why DNS is important: Without it you'd need to memorize IP addresses for every site. It also allows a domain to point to different IPs over time (load balancing, CDN, migration) without users noticing.